

SOFTWARE ENGINEERING – LAB 03

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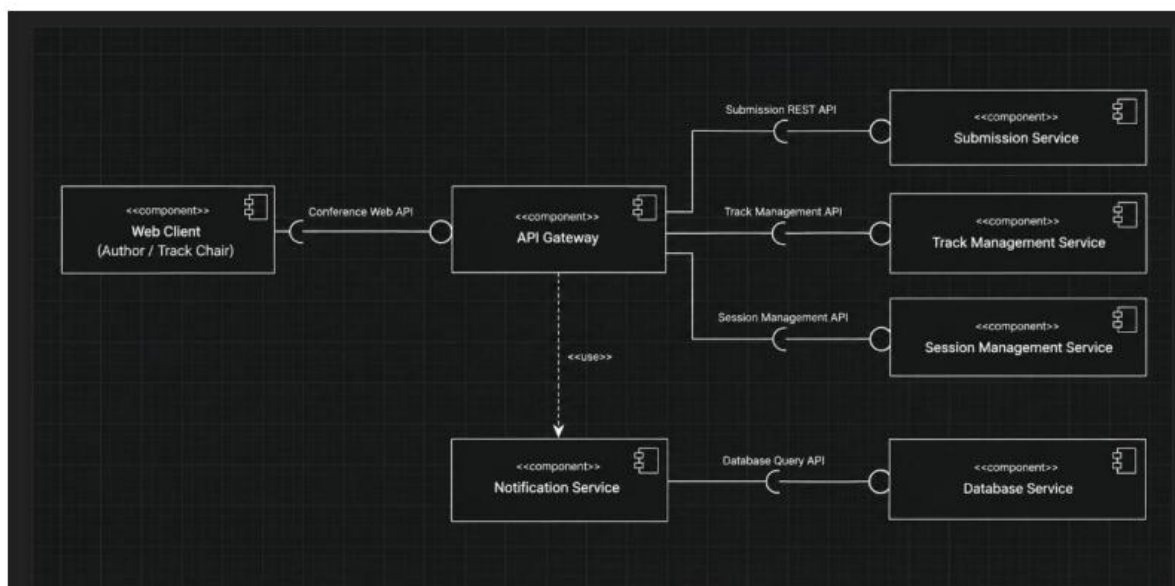
SRN : PES2UG24CS244

SECTION : D

SCENARIO : Virtual Technical Conference Platform

Scenario id - 51

UML DIAGRAM



JUSTIFICATION:

Architecture Selection: Microservices Architecture

We chose **Microservices Architecture** for the **Virtual Technical Conference Platform**. The system can be divided into independent services for different conference activities, allowing each service to perform a specific function without making the entire system dependent on one component.

Reason 1: Microservices allow individual conference services to be **scaled independently**. During a virtual conference, a large number of users may access sessions or conference content simultaneously, while other functions such as submission or track management may have lower demand. Independent scaling helps the platform handle varying workloads efficiently.

Reason 2: Microservices provide **fault isolation**. If one service, such as notifications or session management, experiences a failure, other services can continue operating independently. This improves the reliability and fault tolerance of the conference platform.

Security Advantage: The architecture allows security controls and access restrictions to be applied independently to different services. This supports controlled access to functions such as paper submission and track management, helping protect conference data from unauthorized access.

Performance Benefit: Services experiencing high demand can be scaled independently rather than scaling the entire application. This helps the platform maintain better performance when many participants access conference services concurrently.